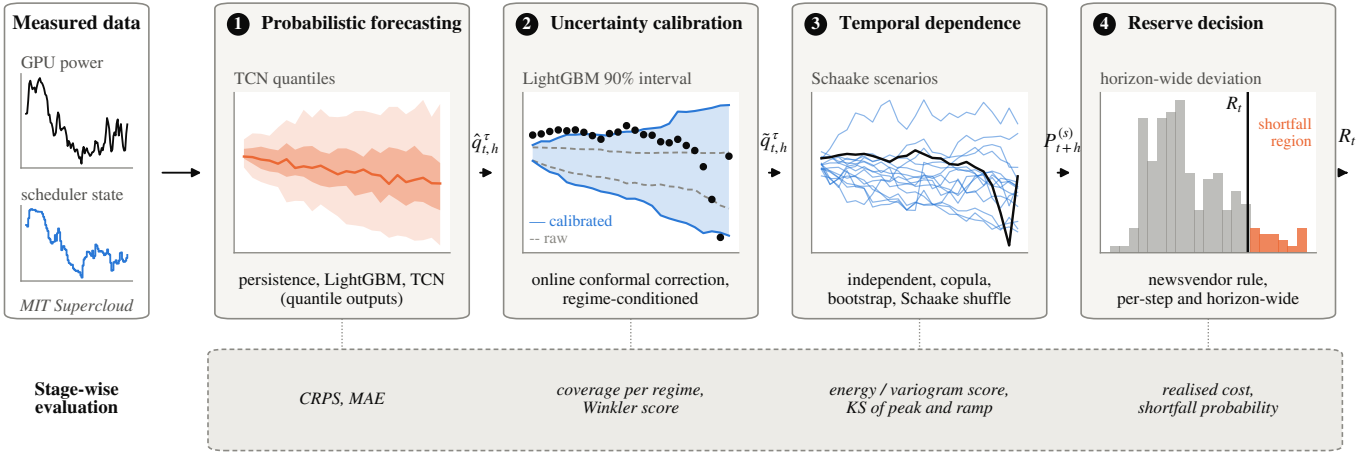
Measured data
GPU power
scheduler state
MIT Supercloud
1 Probabilistic forecasting
TCN quantiles
persistence, LightGBM, TCN
(quantile outputs)
$\hat{q}^{\tau}_{t,h}$
2 Uncertainty calibration
LightGBM 90% interval
calibrated
raw
online conformal correction,
regime-conditioned
$\tilde{q}^{\tau}_{t,h}$
3 Temporal dependence
Schaake scenarios
independent, copula,
bootstrap, Schaake shuffle
$P^{(s)}_{t+h}$
4 Reserve decision
horizon-wide deviation
R_t
shortfall
region
newsvendor rule,
per-step and horizon-wide
R_t
Stage-wise
evaluation
CRPS, MAE
coverage per regime,
Winkler score
energy / variogram score,
KS of peak and ramp
realised cost,
shortfall probability